

Experiment -1

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1. Setting up of the platform

C programming can be practiced online or offline, depending on the available resources. For this experiment, a Linux environment is required.

Online: CoCalc is a browser-based cloud platform that can provide a Linux terminal and programming environment. Official site: <https://cocalc.com/>

Offline: A Linux environment can be installed or run locally on the computer without depending on a browser-based platform.

Methods: dual boot, single boot, a bootable USB drive, or a virtual machine (VM).

1. Virtual Machine Manager (Hypervisor)

Definition: A virtual machine manager, or hypervisor, is software that creates and manages virtual machines by allocating the host computer's processor, memory, storage, and other hardware resources to guest operating systems.

Type-1 (Baremetal)

Definition: A Type-1 (bare-metal) hypervisor runs directly on the physical hardware rather than on a conventional host operating system. It manages the hardware and the guest operating systems.

Examples: VMware ESXi, Microsoft Hyper-V, Xen, and KVM.

Type-2 (Hosted)

Definition: A Type-2 (hosted) hypervisor runs as an application on top of a host operating system such as Windows, Linux, or macOS. The host operating system handles access to the physical hardware.

Examples: Oracle VirtualBox, VMware Workstation, VMware Fusion, and Parallels Desktop.

For this experiment, Oracle VirtualBox is used as the Type-2 hypervisor.

1. Installation of VMM & VM

Step-1: Download Oracle VirtualBox from the official downloads page:

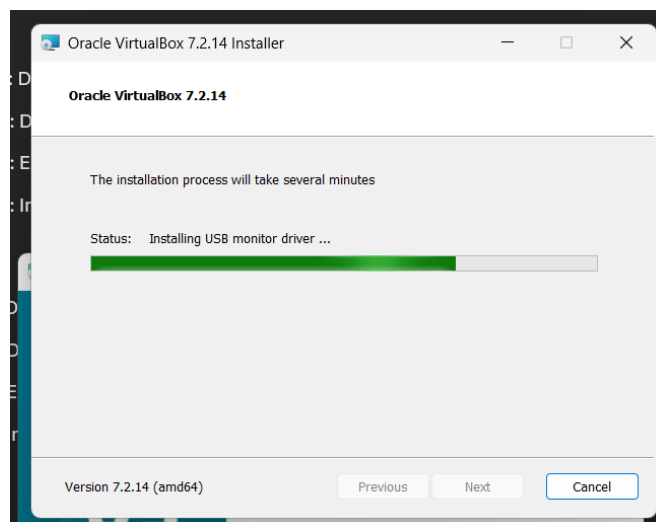
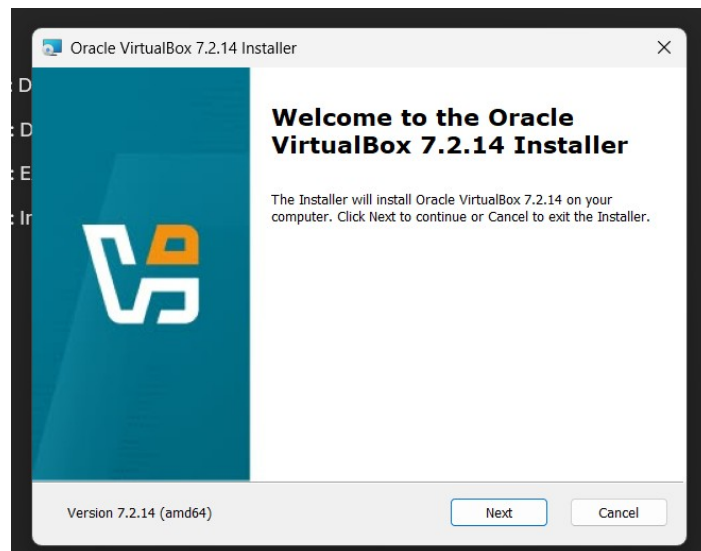
<https://www.virtualbox.org/wiki/Downloads>

Step-2: Download the pre-built Kali Linux image for VirtualBox from:

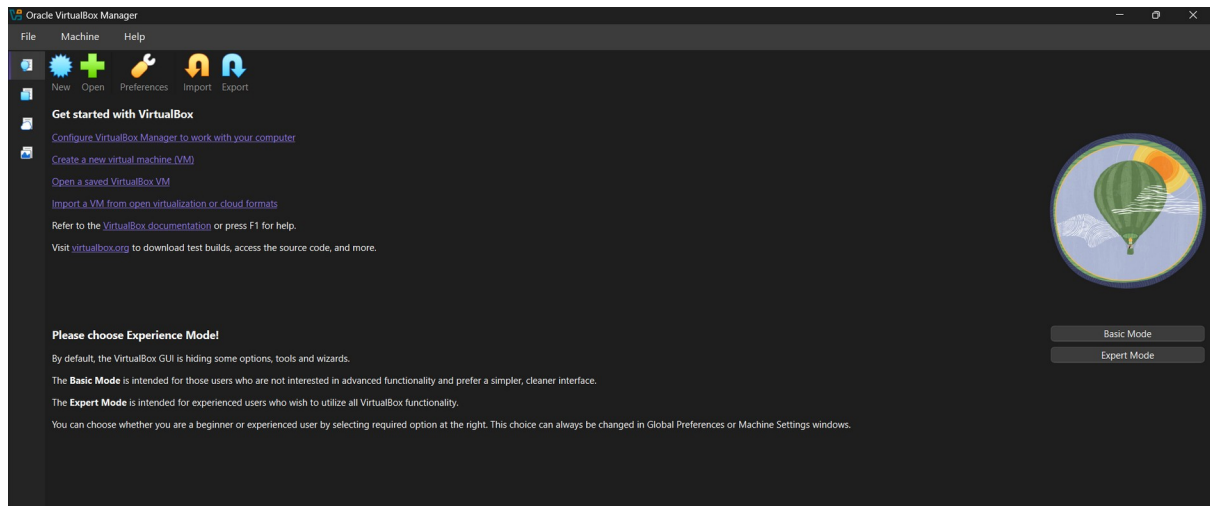
<https://www.kali.org/get-kali/#kali-virtual-machines>

Step-3: Extract the downloaded Kali Linux archive to a suitable folder. The extracted folder contains the VirtualBox virtual-machine files.

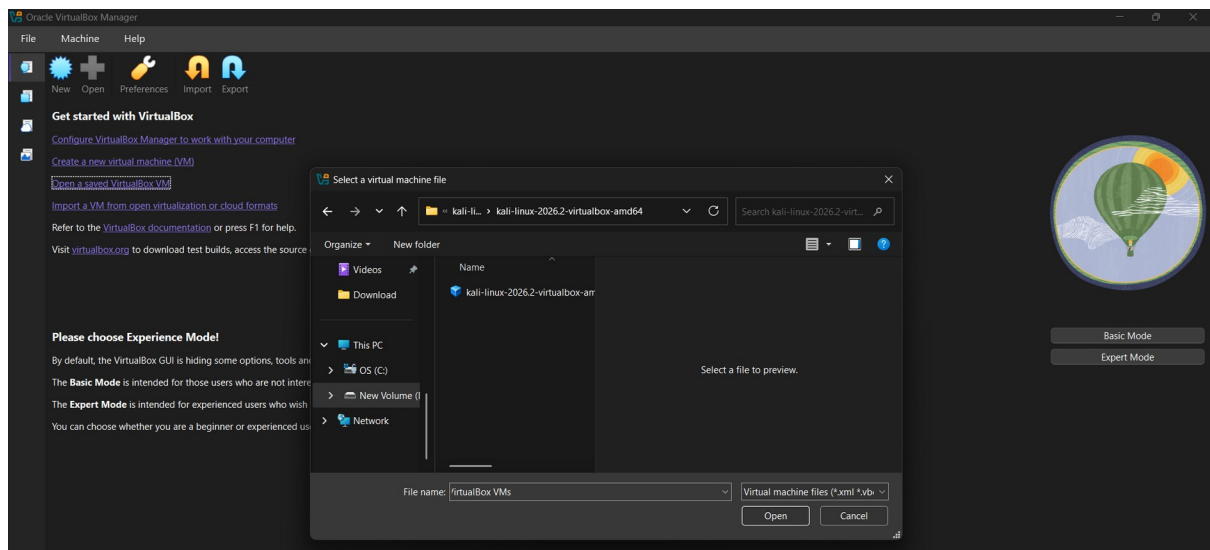
Step-4: Run the Oracle VirtualBox installer and follow the on-screen instructions to finish the installation.

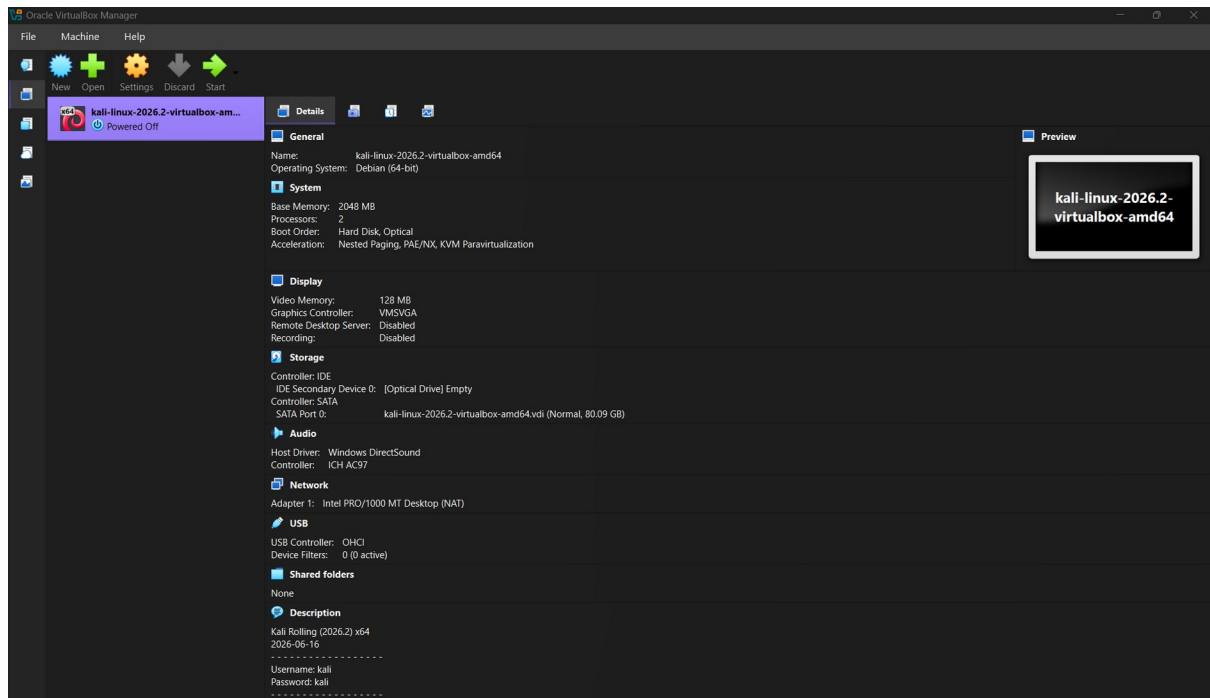


Step-5: Open Oracle VirtualBox Manager.

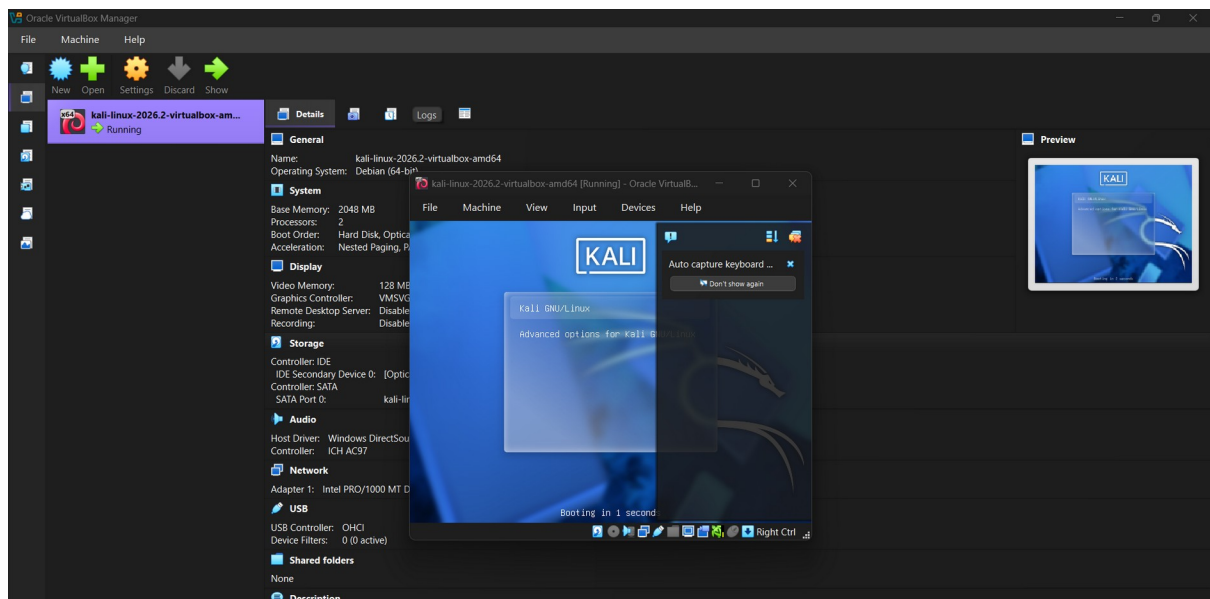


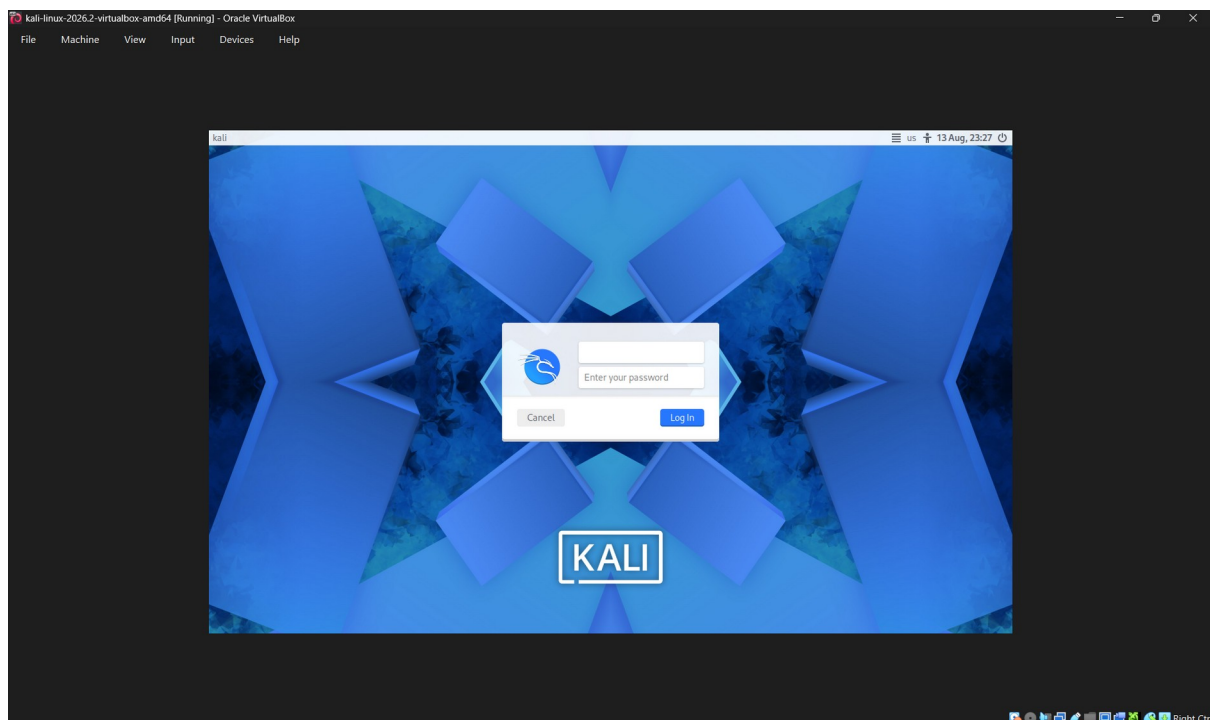
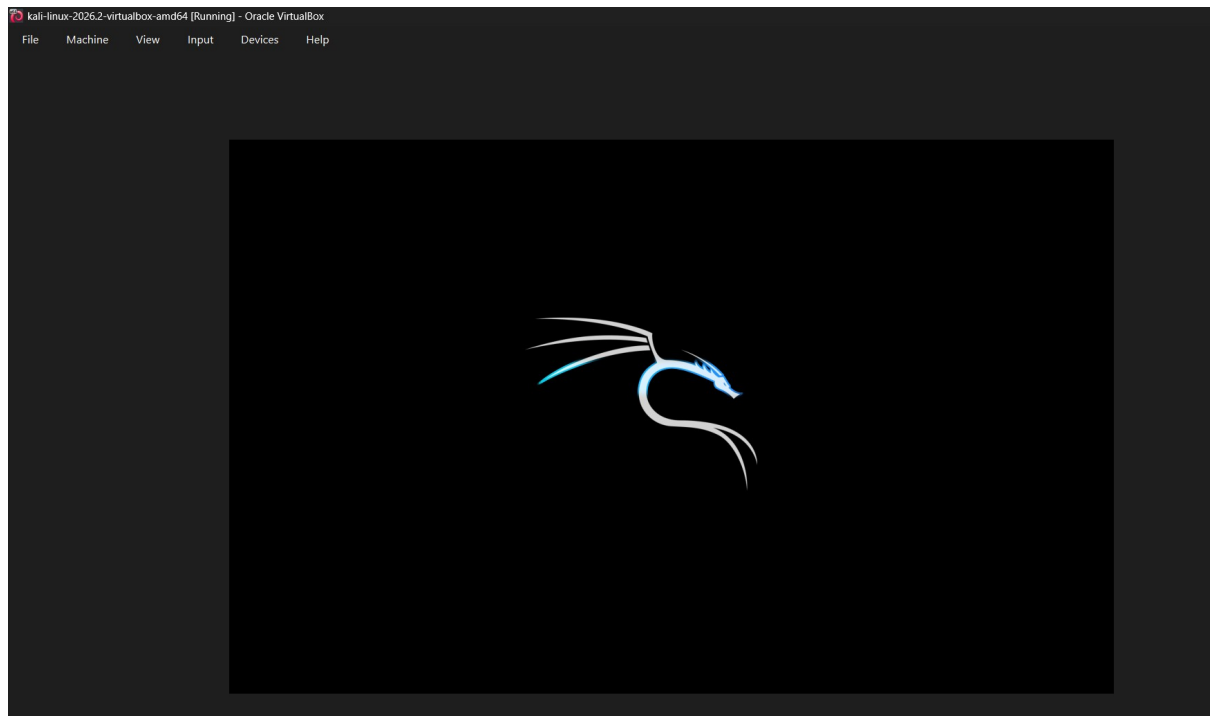
Step-6: Select Open or Add, browse to the extracted Kali Linux .vbox file, and open it to register the saved virtual machine in VirtualBox.





Step-7: Select the Kali Linux virtual machine and click Start. At the login screen, use the default username kali and password kali, and then change the password if instructed.





Result: The Type-2 virtual machine manager (Oracle VirtualBox) and the Kali Linux virtual machine were installed, registered, and started successfully.